Yvonne Wang (Hongyan Wang)

Based in London, UK | no visa constraints

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PROFILE

Computational mechanics researcher (PhD, Imperial College London) with 10+ years of turning physical systems into simulations, data and decisions for aerospace and automotive partners (Airbus, Short Brothers). Extensive experience in building material models, deploying large-scale simulations on HPC, developing Python pipelines that automate the workflow, and applying machine learning and optimisation to engineering problems. Proven record of close collaboration with industrial partners, real-world engineering impact and peer-reviewed publications.

PROFESSIONAL EXPERIENCE

Imperial College London | Short Brothers & Airbus — **London, UK**
Research Associate, Simulation Engineer (ATI project) — Nov. 2024 – present

- **Solver software:** re-architected a legacy Fortran material library (UMAT, USDFLD, VUMAT) into modular, plug-and-play components and optimised the code structure, cutting run time by 60%; built a Python verification and validation (V&V) framework whose single command-line interface (CLI) runs three-tier automated tests (integration point, single element, coupon).
- **Simulation pipeline:** automated the Abaqus workflow in Python, from model generation and high-performance computing (HPC) cluster submission to energy-balance monitoring and post-processing of large-scale FE simulations; built an interactive dashboard to visualise each damage mode at any time step and compare results across designs.
- **Novel bio-inspired laminate design:** designed helicoidal and Double-Double layups with a stiffness-matching optimiser; experiments showed up to 22% higher compression-after-impact (CAI) strength than the baseline. Analysed the FE results in depth to understand the damage mechanisms, which guided later model improvements and design decisions.
- **Multi-fidelity FE modelling:** built (1) low-fidelity models for fast screening of new design concepts and (2) high-fidelity, physics-based models that replicate the actual test conditions and predict the quasi-static and dynamic (impact and post-impact) response of composite structures with novel layups; correlated both with experiments to establish their accuracy and limits.
- **Performance prediction from scarce test data:** built a normalised regression model that predicts the post-impact strength of a new laminate design from physics-derived features (laminate stiffness and FE damage areas grouped by damage mechanism). It predicted the CAI strength of two designs outside the training set to within 6% of the later test results, which removes the need for a CAI simulation of each new design and saves an estimated 60% of simulation time.
- **Collaboration and communication:** worked closely with industrial partners and presented the findings at international conferences (ECCM22, TexComp, ECCOMAS).

University of Southampton | Airbus — **Southampton, UK**
Research Fellow, Digital Design & Data Science Engineer — Nov. 2023 – Nov. 2024

- **Automated cable-routing software:** discretised CAD geometry into 3D graphs of up to 253k nodes (Rhino/Grasshopper), rebuilt the graph structure in Python (NetworkX), and routed cables with shortest-path algorithms (A*, Dijkstra) to minimise cable length while satisfying regulatory and physical constraints (harnessing, functional separation, obstacle avoidance).
- **Geometric path generation:** developed a NURBS interpolation algorithm that turns discrete graph paths into physically realistic 3D cable geometries, enforcing minimum bend radius and inlet/outlet direction constraints.
- **Constraint and packing logic:** solved the smallest-enclosing-circle problem to bundle and unbundle cables dynamically within structural limits; used KD-tree spatial indexing for fast nearest-neighbour queries to enforce separation and structural-affinity rules in real time.
- **Installation-sequence optimisation:** modelled the installation sequence as a constrained travelling salesman problem (TSP) and solved it with permutation-based genetic algorithms (DEAP) combined with heuristics, reducing total system mass by 3.5%.
- **Interactive dashboard:** built a Plotly/Dash dashboard that renders 3D graph networks and cable paths, with a human-in-the-loop interface where users trace constraint compliance and re-compute routes instantly by adjusting the optimisation weights.

Imperial College London — **London, UK**
Ph.D. Researcher, Computational Modelling of Material Processing — Oct. 2018 – Jul. 2023

- **Constitutive modelling:** developed novel constitutive models that capture the stress–strain behaviour of composites across temperatures and strain rates, unifying micro-scale physics, meso-scale stochastic fibre distribution and

macro-scale response so that they apply across material states and compositions; validated experimentally outside the calibration range.

- **ML for microstructure–property linkage:** developed a generalisable framework that predicts the effective properties of composites from their microstructure. Automated in Python the generation of 10,000 random microstructures, their RVE simulations for nine material–temperature combinations (90,000 simulations) and the data extraction. A CNN (PyTorch) combined with a physics-based constitutive model predicted the stress–strain curves of unseen microstructures and materials with ∼3% average error.
- **Process simulation:** implemented the models in Abaqus as VUMAT subroutines; simulated stamp forming of dome-shaped demonstrators to predict manufacturing defects (surface wrinkles, cracks), and used the results to identify forming windows and optimise blank shape, layer blocking and forming temperature.
- **Image processing for data acquisition:** developed a bespoke digital image correlation (DIC) technique with custom MATLAB image-processing algorithms to measure the large strains in thermoplastics that standard DIC could not resolve.
- **Process innovation:** developed induction and resistance heating methods that significantly improved the formability of carbon-fibre thermoplastic composites (CFRTPs), securing Royal Society funding (£20k).

Credit Risk Data Science Project, Lloyds Banking Group (link) UK | Oct. 2023

- Analysed customer loan-repayment behaviour on an 18,000-record dataset: data processing (cleaning, encoding, transformation), visualisation and statistical evaluation. Trained and benchmarked Logistic Regression, Random Forest and XGBoost models to predict the repayment probability of new customers, and compared them by Gini coefficient.

EDUCATION

Imperial College London UK | Oct. 2018 – Jul. 2023
Ph.D. in Mechanical Engineering | full scholarship (∼£140K)

Shanghai Jiao Tong University China | Oct. 2015 – Mar. 2018
M.Sc. in Material Science and Engineering | GPA: 3.7/4.0 (top 10%)

Huazhong University of Science and Technology China | Oct. 2011 – Jun. 2015
B.Sc. in Material Science and Engineering | GPA: 3.9/4.0 (top 2%)

SELECTED PUBLICATIONS, TALKS & PATENTS (Full list in Google Scholar)

[1] Z. Ding, H. R. Attar, H. Wang, et al., "Integrating **convolutional neural network** and constitutive model for rapid prediction of stress-strain curves in fibre reinforced polymers: A generalisable approach." *Materials & Design* 241 (2024): 112849. (DOI)

[2] H. Wang, E. Kazemi, S. T. Pinho, et al., "Design and FEA of bio-inspired carbon fibre-reinforced polymer (CFRP) laminates for improved low-velocity impact (LVI) damage tolerance." The 22nd European Conference on Composite Materials (ECCM22), Oslo, Norway, 2026.

[3] H. Wang, E. Kazemi, S. T. Pinho, "Finite element analysis (FEA) of bio-inspired helicoid carbon fibre-reinforced polymers (CFRPs) under low-velocity impact (LVI)." The 10th ECCOMAS Thematic Conference on the Mechanical Response of Composites, Vienna, Austria, 2025.

[4] H. Wang, Z. Ding, et al., "Experimental characterisation and constitutive modelling of the intra-ply tensile and shear properties of UD FRTPs under solid-state stamp forming conditions." *Composites Part A* 179 (2024): 108034. (DOI)

[5] H. Wang, et al., "Experimental and constitutive modelling studies of semicrystalline thermoplastics under solid-state stamp forming conditions." *Polymer* 228 (2021): 123939. (DOI)

[6] **Patent:** Z. Ding, N. Li, H. Wang, et al., "A method of forming metal alloys and fibre-reinforced polymers in one-shot stamping." P79563GB, 2024.

SKILLS

- **Simulation & physics:** non-linear FEA (Abaqus Standard/Explicit, UMAT/VUMAT), constitutive modelling of polymers and composites, thermo-mechanical coupled modelling.
- **Composite mechanics:** failure and damage modelling (LaRC05), impact and compression after impact, thermo-mechanical behaviour, multi-scale and multi-fidelity modelling, validation against tests (DIC, C-scan, CT).
- **Data science & ML:** Python (NumPy, SciPy, Pandas), scikit-learn, XGBoost; CNN surrogate models, regression with cross-validation, parameter calibration, sensitivity analysis, Monte Carlo sampling, statistics on simulation and test data.
- **Software & HPC:** Python, Fortran, MATLAB, C/C++; HPC clusters, automated simulation pipelines, V&V / regression testing, CLI tools, Git, Plotly/Dash.
- **Optimisation:** constrained and combinatorial optimisation, genetic algorithms (DEAP), graph search (A*, Dijkstra; NetworkX).